

# Sagnik Chatterjee

✉ sagnik.chatterjee@tifr.res.in

✉ chatsagnik@gmail.com

🌐 chatsagnik.github.io

## Education and Employment

- DEC 2025 – Present    📖 **Visiting Fellow (Postdoctoral Researcher)** at STCS, TIFR Mumbai.  
Host: Prof. Jatin Batra.
- Jul 2019 – Jun 2025    📖 **Ph.D.** in Computer Science and Engineering at IIIT-Delhi.  
Advisor: Prof. Debajyoti Bera.
- SEP 2017 – MAR 2019    📖 **Staff Consultant**, Oracle Financial Services Software Limited, Bangalore.
- Aug 2013 – Aug 2017    📖 **B.Tech.** in Computer Science and Engineering at MAKAUT (formerly, WBUT-Inhouse).  
GPA: 8.10/10.

## Research

### Ph.D. Thesis

- 1    **S. Chatterjee**, “Designing quantum learning algorithms for classical objects,” Available at <https://repository.iiitd.edu.in/xmlui/handle/123456789/1758>, PhD thesis, IIIT-Delhi, May 2025.

### Journal Articles

- 1    Y. Saxena, T. Sapv, **S. Chatterjee**, and R. K. Lee, “Generalized cross-resonance scheme for maximally-entangling two-qutrit gates,” *Physical Review Research*, Jun. 2026. 📄 DOI: 10.1103/qqtg-x6jg.
- 2    **S. Chatterjee**, R. Bhatia, P. S. Chani, and D. Bera, “Quantum boosting using domain-partitioning hypotheses,” *Quantum Machine Intelligence*, vol. 5, no. 2, pp. 1–20, 2023. 📄 DOI: 10.1007/S42484-023-00122-3.

### Conference Proceedings

- 1    S. Das<sup>†</sup>, **S. Chatterjee**<sup>†</sup>, and J. Batra, “Convex basins in single-index model loss landscapes: Applications to robust recovery under strong adversarial corruption,” in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. 📄 URL: <https://arxiv.org/abs/2605.29497>.
- 2    M. Mukherjee, **S. Chatterjee**, and A. Sethi, “Perfect secret key generation for a class of hypergraphical sources,” in *IEEE International Symposium on Information Theory (ISIT)*, 2026. 📄 URL: <https://arxiv.org/abs/2601.10697>.
- 3    **S. Chatterjee**<sup>†</sup>, M. Mukherjee<sup>†</sup>, and A. Sethi<sup>†</sup>, “Generalization bounds for dependent data using online-to-batch conversion,” in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. 📄 URL: <https://proceedings.mlr.press/v258/chatterjee25b.html>.
- 4    **S. Chatterjee**, T. SAPV, and D. Bera, “Efficient quantum agnostic improper learning of decision trees,” in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024. 📄 URL: <https://proceedings.mlr.press/v238/chatterjee24a.html>.

<sup>†</sup> denotes equal contribution.

### Under Review

- 1    **S. Chatterjee**, “The Quantum Learning Menagerie (A survey on Quantum learning for Classical concepts),” 2026. 📄 URL: <https://arxiv.org/abs/2602.01054>.

<sup>†</sup> denotes equal contribution.

### Workshop Papers and Posters

- 1    **S. Chatterjee** and V. Kungurtsev, *Quantum solutions to the privacy vs. utility tradeoff*, 2023. arXiv: 2307.03118.
- 2    **S. Chatterjee** and D. Bera, *Applying the quantum alternating operator ansatz to the graph matching problem*, Extended Abstract at the The 20th Asian Quantum Information Science (AQIS) Conference, 2020. arXiv: 2011.11918.

## Teaching

### Instructor (Short Courses)

- Mar 2026  Designed and taught a mini-course (6 lectures) at **IIT (ISM) Dhanbad**.  
**Topic:** axioms of quantum computing, quantum query speedups for promise problems, variational quantum algorithms, and hands-on introduction to QISKIT.
- Feb 2024  **Designed and taught 3 lectures** at the **Chennai Mathematical Institute**.  
**Topic 1:** The Harrow-Hassidim-Lloyd Algorithm and its extensions.  
**Topic 2:** Block Encoding and Linear Combination of Unitaries.
- Aug 2022  Designed and taught 3 lectures for a refresher module at **IIIT-Delhi**.  
**Topic:** C programming for Operating Systems.

### Teaching Assistantship (IIITD)

- Theory of Computation  Winter 2020, Winter 2021, Winter 2024. **[Head TA]**
- Data Structures  Summer 2022. **[Head TA]**
- Intro to Quantum Computing  Winter 2023. **[Sole TA]**
- Modern Algorithm Design  Monsoon 2020, Monsoon 2021. **[Sole TA]**

My duties included designing and conducting tutorials, creating assignments, and holding office hours and remedial sessions for all the above courses.

## Research Visits and Internships

- AUG 2024– SEP 2024  **ACMU, Indian Statistical Institute**, Kolkata.  
**Host:** Prof. Sourav Chakraborty.
- JUL 2023– AUG 2023  **Czech Technical University**, Prague.  
**Host:** Prof. Jakub Marecek, Prof. Vyacheslav Kungurtsev.
- FEB 2017 – APR 2017  **Systems Engineering Intern**, Infosys Limited.
- JUN 2016– JUL 2016  **Data-Science Intern**, AICircle Pte Ltd.

## Invited Talks

- Oct 2025  STCS seminar, TIFR Mumbai.  
*Generalization bounds for dependent data using online-to-batch conversion.*
- May 2025  CS Colloquium, Ashoka University.  
*Efficient quantum agnostic improper learning of decision trees.*
- Mar 2025  Quantum Computing Workshop by Edunautic, IIT Delhi.  
*Full day workshop from fundamentals to hands-on instruction.*
- Jan 2025  Young Scientists Session at QAC 2025 symposium, C-DAC and DIAT Pune.  
*Modern Algorithmic Primitives in Quantum Computing.*
- Oct 2024  ACMU seminar, ISI Kolkata.  
*Generalization bounds for dependent data using online-to-batch conversion.*
- July 2024  Recent Trends in Algorithms Workshop 2024.  
*Efficient quantum agnostic improper learning of decision trees.*
- June 2024  ACMU seminar, ISI Kolkata.  
*Efficient quantum agnostic improper learning of decision trees.*
- February 2024  Quantum Computing Semester, Chennai Mathematical Institute.  
*Quantum Algorithms for Linear Algebra.*  
*Block Encodings and Linear Combination of Unitaries.*
- September 2023  Center for Quantum Computing Science, University of Latvia.  
*Efficient quantum agnostic improper learning of decision trees.*

## Invited Talks (continued)

- IDA Seminar, Czech Technical University.  
*Efficient quantum agnostic improper learning of decision trees.*
- November 2022 ■ Short Talk at the 6th Quantum Techniques in Machine Learning (QTML 2022).  
*Quantum boosting using domain-partitioning hypotheses.*
- March 2022 ■ Theory Seminar, Indian Institute of Information Technology Delhi.  
*Quantum boosting using domain-partitioning hypotheses.*
- December 2020 ■ Faculty Development Programme, JNTU Anantapur.  
*Quantum Machine Learning.*

## Miscellaneous

### Awards and Achievements

- MAY 2026 ■ Recognized as a **Gold Reviewer** at **ICML 2026**.
- FEB 2026 ■ Invited poster at **ACM ARCS 2026**.
- FEB 2025 ■ Invited poster at **ACM ARCS 2025**. (Declined)
- AUG 2024 ■ **IIITD Dean's List** for Best Teaching Assistant (Theory of Computation).
- APR 2024 ■ AISTATS 2024 Registration Grant.
- JAN 2022 ■ QIP 2022 Student Travel Award. (Not availed due to COVID restrictions)
- FEB 2020 ■ **Runners Up, IBMQ Awards** in the *Teach Me Quantum 2019* category. (4000 USD cash prize)
- OCT 2019 ■ Full Scholarship to attend the 4th Winter School in CSE organized by the IIAS, HUJI, Jerusalem.

### Reviewing

- Conference ■ **STACS** ('25); **ICML** ('25,'26); **NeurIPS** ('24, '25); **ICLR** ('25,'26); **AISTATS** ('25,'26).
- Journal ■ **Scientific Reports** ('24); **Quantum** ('24, '25).

### Mentoring

- DEC 2025 – Present ■ Santanu Das. PhD candidate at TIFR Mumbai.
- JAN 2024 – MAY 2025 ■ Alhad Sethi. B.Tech student at IIIT Delhi.
- JAN 2021 – JUL 2023 ■ Parmeet Singh Chani. B.Tech student at DTU, Delhi.
- Rohan Bhatia. B.Tech student at DTU, Delhi.

### Workshop

- Workshops ■ Co-organiser and Webadmin for the QISE workshop at FSTTCS 2021.
- Seminars (IIIT-D) ■ Talks on Quantum Computing, Spectral Graph Theory, Theory Reading Group, Ketchup talks.

## References

- Prof Debajyoti Bera ■ Associate Professor,  
Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi),  
✉ dbera@iiitd.ac.in
- Prof Jatin Batra ■ Reader (Assistant Professor),  
Tata Institute of Fundamental Research, Mumbai (TIFR-Mumbai),  
✉ jatin.batra@tifr.res.in
- Prof Manuj Mukherjee ■ Assistant Professor,  
Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi),  
✉ manuj@iiitd.ac.in